Soumen Kundu

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Education

Calcutta University **2016 – 2020**
Bachelor of Science (B.Sc - General) — Completed *Kolkata, West Bengal*

Indira Gandhi National Open University **Jul. 2022 – Jun. 2025**
Master of Computer Applications (MCA) — Completed *Kolkata, West Bengal*

Professional Summary

AI/ML Engineer with 4+ years of experience in healthcare, building production-grade ML and Generative AI systems. Specialized in LLM orchestration, multi-agent architectures, and RAG pipelines with LangGraph and LangChain. Experienced in deploying scalable AI systems with FastAPI, Docker, Kubernetes, and AWS, with a focus on performance, observability, and real-world impact.

Experience

Charnock Hospital Pvt. Ltd. (Multiple Units) **Sep 2021 – Present**
IT Executive (Automation, Data, AI Systems) *Kolkata, West Bengal*

- Automated data pipelines using Python and SQL, reducing manual reporting effort.
- Performed data analysis to support operational decision-making across hospital units.
- Developed internal tools, softwares & websites and dashboards improving reporting efficiency.
- Managed and maintained enterprise systems ensuring high availability.

Projects

AI Blog Writing Agent (Multi-Agent GenAI System) | *LangGraph, FastAPI, Streamlit, Gemini, Tavily*

- Built a **multi-agent LLM system using LangGraph**, enabling automated blog generation with parallel execution across tasks.
- Implemented **RAG pipeline using Tavily**, improving factual accuracy of generated content.
- Developed FastAPI + Streamlit application with Gemini-based image generation and **LangSmith** tracing for observability.

Vehicle Insurance Prediction – End-to-End MLOps Project | *Python, Scikit-learn, FastAPI, Docker, AWS, MLflow, DVC*

- Designed and implemented a **production-grade ML pipeline** covering data ingestion, exploratory data analysis, feature engineering, model training, evaluation, and inference.
- Built REST APIs using **FastAPI** and deployed models via **Docker containers on AWS**.
- Implemented **CI/CD pipelines with GitHub Actions** for automated testing and deployment.
- Tracked experiments using **MLflow** and versioned datasets/models with **DVC**.
- Used **MongoDB Atlas** for scalable data storage.

Sentiment Analysis End-to-End MLOps Pipeline | *Python, NLP, Scikit-learn,Docker, Kubernetes,AWS, Prometheus, Grafana*

- Developed an NLP sentiment analysis pipeline using **TF-IDF feature extraction and classical ML models** for text classification.
- Automated training and deployment using **Docker and GitHub Actions**.
- Deployed services on **AWS EKS (Kubernetes)** with real-time monitoring.
- Implemented **Prometheus and Grafana dashboards** for model and system metrics.

Pneumonia Detection from Chest X-Ray Images | *Python, TensorFlow, Keras, CNN, OpenCV, Streamlit, Hugging Face*

- Built a **CNN-based deep learning model** for pneumonia detection from X-ray images.
- Optimized image preprocessing and inference for real-time prediction.
- Deployed the model as a **Streamlit web application**.
- Managed large model artifacts using **Hugging Face Hub**.

Multiple Disease Prediction App | *Python, Scikit-learn, Streamlit, NumPy, Pandas*

- Built a multi-model ML web application predicting diseases such as diabetes, heart disease, Parkinson's, kidney disease, and breast cancer.
- Integrated multiple trained classification models into a single Streamlit application.
- Enabled real-time predictions using user-provided medical parameters.
- Demonstrated complete ML workflow including model training, serialization, and deployment.

Technical Skills

Languages: Python, SQL, C
Data Analysis & Visualization: Pandas, NumPy, Exploratory Data Analysis (EDA), Feature Engineering, Matplotlib
Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, CNN, NLP (TF-IDF)
MLOps: MLflow, DVC, Docker, GitHub Actions, CI/CD, Experiment Tracking, Model Versioning, Model Deployment
Cloud & DevOps: AWS (S3, EC2, ECR, EKS, Bedrock), Docker, Kubernetes, CI/CD Pipelines, Model Deployment, Scalable Infrastructure, Monitoring & Observability
Databases: MySQL, MongoDB Atlas
Monitoring: Prometheus, Grafana
Web & APIs: FastAPI, Flask, REST APIs, Streamlit
Generative AI, LLMs: RAG, Self-RAG, Corrective RAG (CRAG), LangChain, LangGraph, LangSmith, AI Agents, Multi-Agent Systems, Prompt Engineering, LLM Evaluation, Embeddings, Vector Databases (FAISS, Pinecone), MCP, OpenAI APIs, Gemini APIs, Claude Code, LoRA, QLoRA, PEFT, Hugging Face Transformers, Fine-Tuning & Inference Optimization, LLM Workflows, LLMOps, AI System Design
System Design & Distributed Systems:Client-Server Architecture, API Gateway, Horizontal vs Vertical Scaling, Load Balancing, Caching (Redis), CDN, Database Replication, Database Sharding, Consistent Hashing, CAP Theorem, Message Queues (Kafka), Object Storage, Idempotency, Long Polling / Short Polling, Microservices Architecture, High Availability & Fault Tolerance
Tools: Git, Linux, Hugging Face Hub